THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. When a river dries and the land becomes bare, the soil is more likely to: 1

(a) erode **(b)** become fertile
(c) absorb more water **(d)** grow plants

2. If a satellite takes 90 minutes for one orbit, the number of orbits in 24 hours is: 1

(a) 90 **(b)** 16 **(c)** 8 **(d)** 24

3. Which of the following correctly orders the steps of a municipal water treatment? 1

(a) sedimentation, filtration, disinfection **(b)** boiling, magnet
(c) disinfection, filtration **(d)** filtration, sedimentation

4. Which characteristic makes 'excretion' essential for life? 1

(a) it makes food **(b)** accumulated wastes are toxic
(c) it helps in growth **(d)** it makes light

5. A village uses a solar pump instead of a diesel pump. The main benefit is: 1

(a) more smoke **(b)** no fuel cost and no smoke
(c) more fuel **(d)** more noise

6. How would the length of day change if the Earth rotated twice as fast? 1

(a) stay the same **(b)** double **(c)** halve **(d)** vanish

7. A mixture of two liquids that do not mix is separated by: 1

(a) sieving **(b)** winnowing
(c) filtration through paper **(d)** decantation after the layers form

8. Which of these cannot be taken as proof of life when used alone? 1

(a) movement **(b)** reproduction
(c) respiration **(d)** growth from inside

9. A hill slope is turned into steps and trees planted. The main effect is: 1

(a) no rain **(b)** less soil washed away
(c) higher hill **(d)** more soil lost

10. On a full-moon night the Moon rises roughly at: 1

(a) sunrise **(b)** sunset **(c)** midnight **(d)** noon

11. A mixture of sand and iron in 1:1 by mass weighing 40 g. After using a magnet, the iron obtained should be: 1

(a) 40 g **(b)** 10 g **(c)** 0 g **(d)** 20 g

12. A calf grows into a cow. Its growth involves: 1

(a) melting **(b)** growth from within using food

(c) absorption of sand **(d)** addition of stones

13. **Assertion (A):** Coal is a non-renewable resource. 1

Reason (R): Nitrogen is the most abundant gas in air.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The Moon does not produce its own light. 1

Reason (R): We see the Moon because it reflects sunlight.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** Salt dissolved in water can be separated by filtration. 1

Reason (R): The particles of dissolved salt are too small and pass through filter paper.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. How do earthworms and bacteria keep soil fertile? 2

17. Why is a launch near the equator more efficient for satellites? 2

18. Why does tea powder pass through a sieve but not through a filter paper in tea? 2

19. Why do biologists use several characteristics rather than one to define life? 2

20. Why does the quality of soil decrease if the same crop is grown again and again? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A car consumes 1 L of petrol per 15 km. How much petrol will be saved in a month if a person walks or cycles instead of using the car for 60 km? 3

22. Moon takes about 27 days to orbit the Earth. How many orbits in 1 year (365 days)? 3

23. Why is a cloth filter used when preparing curd or paneer? 3

24. A potted plant placed sideways soon curves upward. What does this show? 3

OR

Explain why the burning of plastic is worse than burying it in terms of air quality. 3

25. Why do some comets appear only once in many years? 3

OR

A mixture of oil and sand is given. How will you separate? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. A school in Dehradun carried out an activity on air. Students blew into a balloon, burnt a candle under a glass jar and saw it go out after some time, and noticed that moving air turned a small paper windmill. The teacher said air is a mixture of gases, mostly nitrogen and oxygen. 5

(i) Name the most abundant gas in air.

(ii) Which gas is needed for burning and respiration?

(iii) What is moving air called?

(iv) Why did the candle go out in the closed jar?

27. Analyse the case: A town cuts nearby hills for stone, building and farmland. Predict five consequences. 5

OR

A student records the Moon's shape for 30 days and finds it was a thin crescent on day 3, half on day 8, full on day 15, half on day 22 and none on day 29. Analyse the pattern. 5

END OF QUESTION PAPER — Total: 50 Marks

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